

WeightedCorrelations_H2A-MHC

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```
library(tidyverse)
library(readxl)
library(knitr)
# Data:
H2A <- read_csv("~/internship/workspace/Written Datasets/H2AWorkers_County_Clean_2.csv") # Worker densi
MHC <- read_csv("~/internship/workspace/Data/migrant_health_centers_ncfh.csv") # Migrant health centers
ENV <- read_csv("~/internship/workspace/Written Datasets/FEMA_Disasters_NumEventsGreaterThanZero.csv")
AGO <- read_csv("~/internship/workspace/Written Datasets/AgrCensus_Output_Clean.csv") # Agricultural ou
JUL23 <- read_csv("~/internship/HICAHS/Data/Heat_Ag_HumanRisk/CountyMaxTemp_JUL23.csv") # Maximum tempe
AUG23 <- read_csv("~/internship/HICAHS/Data/Heat_Ag_HumanRisk/CountyMaxTemp_AUG23.csv") # Maximum tempe

# Healthcare facilities in each state
# (Cannot bind due to differing number of columns)
COH <- read_excel("~/internship/workspace/Health Facility Data/ColoradoHealth24_xlsx.xlsx", sheet = 2)
WYH <- read_excel("~/internship/workspace/Health Facility Data/WyomingHealth24_xlsx.xlsx", sheet = 2)
SDH <- read_excel("~/internship/workspace/Health Facility Data/SouthDakotaHealth24_xlsx.xlsx", sheet = 2)
NDH <- read_excel("~/internship/workspace/Health Facility Data/NorthDakotaHealth24_xlsx.xlsx", sheet = 2)
UTH <- read_excel("~/internship/workspace/Health Facility Data/UtahHealth24_xlsx.xlsx", sheet = 2)
MTH <- read_excel("~/internship/workspace/Health Facility Data/MontanaHealth24_xlsx.xlsx", sheet = 1)

# Empty vectors for later
response <- character()
predictor <- character()

# Cleaning Cleaning Cleaning
ENV$county[ENV$county == "lamoure"] <- "la moure"
ENV <- ENV |> select(-matches("coastal|hurricane|tsunami|volcanic", ignore.case = TRUE))
AGO[] <- lapply(AGO, function(col) gsub(",", "", col))

WYH$county <- gsub(" County", "", WYH$county)
MTH$county <- str_to_title(MTH$county)

COH$state <- str_to_title(COH$state)
WYH$state <- str_to_title(WYH$state)
SDH$state <- str_to_title(SDH$state)
NDH$state <- str_to_title(NDH$state)
UTH$state <- str_to_title(UTH$state)
MTH$state <- str_to_title(MTH$state)

JUL23$Name <- gsub(" County", "", JUL23$Name)
AUG23$Name <- gsub(" County", "", AUG23$Name)
```



```

JOIN <- JOIN |> mutate(`CriticalAccessHospitals` = coalesce(`Critical Access Hospitals`, Critical_Access_Hospitals))
JOIN <- JOIN |> select(-`Critical Access Hospitals`, -Critical_Access_Hospitals)

JOIN$county[JOIN$county == "lamoure"] <- "la moure"

#JOIN$H2A_workers[is.na(JOIN$H2A_workers)] <- 0

# JOIN <- JOIN |>
#   group_by(state) |>
#   mutate(H2AStateTotal = sum(H2A_workers, na.rm = TRUE))
#-----
JOIN_CUT <- JOIN |>
  select(
    "state",
    "county",
    "Population2020",
    "BuildingValue",
    "AgricultureValue",
    "Areasqmi",
    "H2A_workers",
    "JOIN_CUT$latitude",
    "JOIN_CUT$longitude",
    grep(
      "income|sales|farm_sales|income_net_|commodity_totals|crop_totals_sales|drought|wildfire|heat|heat_index",
      colnames(JOIN),
      ignore.case = TRUE
    ),
    349:length(JOIN)
  )

# Very simplified subset of risk variables
JOIN_CUT <- JOIN_CUT |> select(-grep("Expected|wheat|cattle|hogs|chickens|receipts|historic|index", colnames(JOIN_CUT)))
# Filling NAs
JOIN_CUT[sapply(JOIN_CUT, is.numeric)] <- lapply(JOIN_CUT[sapply(JOIN_CUT, is.numeric)], function(x) ifelse(is.na(x), 0, x))
# Deleting empty columns
JOIN_CUT <- JOIN_CUT[, colSums(!is.na(JOIN_CUT)) > 0]

JOIN_CUT <- JOIN_CUT |>
  left_join(JUL23 |> select(state, county, Jul2023_MaxTemp, `July1901-2000Mean`), by = c("state", "county"))
  left_join(AUG23 |> select(state, county, Aug2023_MaxTemp, `Aug1901-2000Mean`), by = c("state", "county"))

JOIN_CUT <- JOIN_CUT |> arrange(state, county)

cx <- ux <- nx <- mx <- data.frame(state = character(),
                                   correlation = numeric(),
                                   weightedCorrelation = numeric())

COLORADO_MHC <- JOIN_CUT |> filter(state=="Colorado") |> arrange(desc(H2A_workers))

```

```

x <- round(cor(COLORADO_MHC$H2A_workers, COLORADO_MHC$MigrantHealthCenters), 3)
y <- round(cor(COLORADO_MHC$H2A_workers/COLORADO_MHC$Areasqmi, COLORADO_MHC$MigrantHealthCenters/COLORADO_MHC$Areasqmi), 3)
cx <- rbind(cx, data.frame(state = "Colorado", correlation = x, weightedCorrelation = y))

UTAH_MHC <- JOIN_CUT |> filter(state=="Utah") |> arrange(desc(H2A_workers))
x <- round(cor(UTAH_MHC$H2A_workers, UTAH_MHC$MigrantHealthCenters), 3)
y <- round(cor(UTAH_MHC$H2A_workers/UTAH_MHC$Areasqmi, UTAH_MHC$MigrantHealthCenters/UTAH_MHC$Areasqmi), 3)
ux <- rbind(ux, data.frame(state = "Utah", correlation = x, weightedCorrelation = y))

NORTHDAKOTA_MHC <- JOIN_CUT |> filter(state=="North Dakota") |> arrange(desc(H2A_workers))
x <- round(cor(NORTHDAKOTA_MHC$H2A_workers, NORTHDAKOTA_MHC$MigrantHealthCenters), 3)
y <- round(cor(NORTHDAKOTA_MHC$H2A_workers/NORTHDAKOTA_MHC$Areasqmi, NORTHDAKOTA_MHC$MigrantHealthCenters/NORTHDAKOTA_MHC$Areasqmi), 3)
nx <- rbind(nx, data.frame(state = "North Dakota", correlation = x, weightedCorrelation = y))

MONTANA_MHC <- JOIN_CUT |> filter(state=="Montana") |> arrange(desc(H2A_workers))
x <- round(cor(MONTANA_MHC$H2A_workers, MONTANA_MHC$MigrantHealthCenters), 3)
y <- round(cor(MONTANA_MHC$H2A_workers/MONTANA_MHC$Areasqmi, MONTANA_MHC$MigrantHealthCenters/MONTANA_MHC$Areasqmi), 3)
mx <- rbind(mx, data.frame(state = "Montana", correlation = x, weightedCorrelation = y))

state_corr_frame <- rbind(cx, mx, nx, ux)
# state_corr_frame <- rbind(state_corr_frame, data.frame(state = "South Dakota", correlation = NA, weightedCorrelation = NA))
# state_corr_frame <- rbind(state_corr_frame, data.frame(state = "Wyoming", correlation = NA, weightedCorrelation = NA))

colnames(state_corr_frame) <- c("State", "Correlation", "Correlation-Weighted for Sq. Miles")
state_corr_frame <- state_corr_frame |> arrange(desc(Correlation))

# write.csv(state_corr_frame, file = "~/internship/workspace/Written Datasets/H2AWorkers_MHC_WeightedCorrelation.csv")

# Combining H-2A population and migrant health center counts
H2AMHC <- H2A |> left_join(MHC_sum, by = c("state", "county"))
H2AMHC$MigrantHealthCenters[is.na(H2AMHC$MigrantHealthCenters)] <- 0 # replace NAs
Z <- JOIN_CUT |> select(state, county, Areasqmi)
H2AMHC <- H2AMHC |> left_join(Z, by = c("state", "county"))

H2AMHC$Areasqmi[H2AMHC$state == "Montana" & H2AMHC$county == "Fremont"] <- 9266
H2AMHC <- na.omit(H2AMHC)

# Calculate correlation
stx <- data.frame(State = character(),
                  Correlation = numeric(),
                  weightedCorrelation = numeric() )

x <- round(cor(H2AMHC$H2A_workers, H2AMHC$MigrantHealthCenters), 3)
y <- round(cor(H2AMHC$H2A_workers/H2AMHC$Areasqmi, H2AMHC$MigrantHealthCenters/H2AMHC$Areasqmi), 3)

stx <- rbind(stx, data.frame(state = "ALL States", correlation = x, weightedCorrelation = y))
colnames(stx) <- colnames(state_corr_frame)
state_corr_frame <- rbind(stx, state_corr_frame)

colnames(state_corr_frame) <- c("State", "Rho", "Rho: Per Sq. Miles")
kable(state_corr_frame, align = c("r", "c", "c"))

```

State	Rho	Rho: Per Sq. Miles
ALL States	0.289	0.274
North Dakota	0.473	0.486
Montana	0.359	0.258
Colorado	0.325	0.295
Utah	0.196	0.355

```
FinalData <- read_csv("~/internship/workspace/Written Datasets/CombinedData_H2AWorkers_FEMA_AgrOutput_H")
rediTable2 <- data.frame(matrix(ncol = 7, nrow = 3))
colnames(rediTable2) <- c("Exposure", "Colorado", "Utah", "Montana", "North Dakota", "South Dakota", "W")

rediTable3 <- data.frame(matrix(ncol = 7, nrow = 3))
colnames(rediTable3) <- c("Exposure", "Colorado", "Utah", "Montana", "North Dakota", "South Dakota", "W")

FinalData_1 <- FinalData |>
  select(state, county,
         H2A_workers, H2AStateTotal,
         MigrantHealthCenters, Population2020, Areasqmi,
         HeatWaveExposureTotal, HeatWaveExposurePopulation,
         DroughtExposureTotal,
         WildfireExposureTotal, WildfireExposurePopulation)

FinalData_colorado <- FinalData_1 |> filter(state == "Colorado" | state == "colorado")
FinalData_wyoming <- FinalData_1 |> filter(state == "Wyoming" | state == "wyoming")
FinalData_utah <- FinalData_1 |> filter(state == "Utah" | state == "utah")
FinalData_montana <- FinalData_1 |> filter(state == "Montana" | state == "montana")
FinalData_northDakota <- FinalData_1 |> filter(state == "North Dakota" | state == "north dakota")
FinalData_southDakota <- FinalData_1 |> filter(state == "South Dakota" | state == "south dakota")

rediCorrFunc <- function(dataset, response, predictor) {
  c <- round(cor(dataset[[response]], dataset[[predictor]]), 3)
  return(c)
}

# corr w/ H-2A population
rediTable2[1, 2] <- rediCorrFunc(FinalData_colorado, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 2] <- rediCorrFunc(FinalData_colorado, "H2A_workers", "DroughtExposureTotal")
rediTable2[3, 2] <- rediCorrFunc(FinalData_colorado, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 3] <- rediCorrFunc(FinalData_utah, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 3] <- rediCorrFunc(FinalData_utah, "H2A_workers", "DroughtExposureTotal")
rediTable2[3, 3] <- rediCorrFunc(FinalData_utah, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 4] <- rediCorrFunc(FinalData_montana, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 4] <- rediCorrFunc(FinalData_montana, "H2A_workers", "DroughtExposureTotal")
rediTable2[3, 4] <- rediCorrFunc(FinalData_montana, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 5] <- rediCorrFunc(FinalData_northDakota, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 5] <- rediCorrFunc(FinalData_northDakota, "H2A_workers", "DroughtExposureTotal")
rediTable2[3, 5] <- rediCorrFunc(FinalData_northDakota, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 6] <- rediCorrFunc(FinalData_southDakota, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 6] <- rediCorrFunc(FinalData_southDakota, "H2A_workers", "DroughtExposureTotal")
```

```

rediTable2[3, 6] <- rediCorrFunc(FinalData_southDakota, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 7] <- rediCorrFunc(FinalData_wyoming, "H2A_workers", "WildfireExposureTotal")
rediTable2[2, 7] <- rediCorrFunc(FinalData_wyoming, "H2A_workers", "DroughtExposureTotal")
rediTable2[3, 7] <- rediCorrFunc(FinalData_wyoming, "H2A_workers", "HeatWaveExposureTotal")

rediTable2[1, 1] <- "Total Wildfire"
rediTable2[2, 1] <- "Total Drought"
rediTable2[3, 1] <- "Total Heat Wave"

kable(rediTable2, caption = "Correlation Between H-2A Worker Population and Wildfire/Drought/Heatwave")

```

Table 2: Correlation Between H-2A Worker Population and Wildfire/Drought/Heatwave

Exposure	Colorado	Utah	Montana	North Dakota	South Dakota	Wyoming
Total Wildfire	0.189	0.703	0.195	0.191	-0.073	-0.199
Total Drought	0.318	0.652	0.379	0.673	0.177	0.008
Total Heat Wave	0.570	0.591	0.196	0.245	0.133	-0.033

```

# write.xlsx(rediTable2, file = "~/internship/workspace/Written Datasets/rediTable3.xlsx")

# correlation w/ MHCs
rediTable3[1, 2] <- rediCorrFunc(FinalData_colorado, "MigrantHealthCenters", "WildfireExposureTotal")
rediTable3[2, 2] <- rediCorrFunc(FinalData_colorado, "MigrantHealthCenters", "WildfireExposureTotal")
rediTable3[3, 2] <- rediCorrFunc(FinalData_colorado, "MigrantHealthCenters", "WildfireExposureTotal")

rediTable3[1, 3] <- rediCorrFunc(FinalData_utah, "MigrantHealthCenters", "WildfireExposureTotal")
rediTable3[2, 3] <- rediCorrFunc(FinalData_utah, "MigrantHealthCenters", "DroughtExposureTotal")
rediTable3[3, 3] <- rediCorrFunc(FinalData_utah, "MigrantHealthCenters", "HeatWaveExposureTotal")

rediTable3[1, 4] <- rediCorrFunc(FinalData_montana, "MigrantHealthCenters", "WildfireExposureTotal")
rediTable3[2, 4] <- rediCorrFunc(FinalData_montana, "MigrantHealthCenters", "DroughtExposureTotal")
rediTable3[3, 4] <- rediCorrFunc(FinalData_montana, "MigrantHealthCenters", "HeatWaveExposureTotal")

rediTable3[1, 5] <- rediCorrFunc(FinalData_northDakota, "MigrantHealthCenters", "WildfireExposureTotal")
rediTable3[2, 5] <- rediCorrFunc(FinalData_northDakota, "MigrantHealthCenters", "DroughtExposureTotal")
rediTable3[3, 5] <- rediCorrFunc(FinalData_northDakota, "MigrantHealthCenters", "HeatWaveExposureTotal")

rediTable3[1, 6] <- rediCorrFunc(FinalData_southDakota, "MigrantHealthCenters", "WildfireExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero
rediTable3[2, 6] <- rediCorrFunc(FinalData_southDakota, "MigrantHealthCenters", "DroughtExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero
rediTable3[3, 6] <- rediCorrFunc(FinalData_southDakota, "MigrantHealthCenters", "HeatWaveExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero

```

```

rediTable3[1, 7] <- rediCorrFunct(FinalData_wyoming, "MigrantHealthCenters", "WildfireExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero
rediTable3[2, 7] <- rediCorrFunct(FinalData_wyoming, "MigrantHealthCenters", "DroughtExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero
rediTable3[3, 7] <- rediCorrFunct(FinalData_wyoming, "MigrantHealthCenters", "HeatWaveExposureTotal")

## Warning in cor(dataset[[response]], dataset[[predictor]]): the standard
## deviation is zero
rediTable3[1, 1] <- "Total Wildfire"
rediTable3[2, 1] <- "Total Drought"
rediTable3[3, 1] <- "Total Heat Wave"

kable(rediTable3, caption = "Correlation Between MHCs and Wildfire/Drought/Heatwave")

```

Table 3: Correlation Between MHCs and Wild-fire/Drought/Heatwave

Exposure	Colorado	Utah	Montana	North Dakota	South Dakota	Wyoming
Total Wildfire	-0.018	0.386	0.533	-0.051	NA	NA
Total Drought	-0.018	0.124	0.125	0.355	NA	NA
Total Heat Wave	-0.018	0.677	0.243	-0.020	NA	NA

```

# write.xlsx(rediTable3, file = "~/internship/workspace/Written Datasets/rediTable3.xlsx")

```